

Assignment 3: Data Exploration

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Exploration.

Directions

1. Rename this file `<FirstLast>_A03_DataExploration.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Assign a useful **name to each code chunk** and include ample **comments** with your code.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.
7. After Knitting, submit the completed exercise (PDF file) to the dropbox in Canvas.

TIP: If your code extends past the page when knit, tidy your code by manually inserting line breaks.

TIP: If your code fails to knit, check that no `install.packages()` or `View()` commands exist in your code.

Set up your R session

1. Load necessary packages (tidyverse, lubridate, here), check your current working directory and upload two datasets: the ECOTOX neonicotinoid dataset (`ECOTOX_Neonicotinoids_Insects_raw.csv`) and the Niwot Ridge NEON dataset for litter and woody debris (`NEON_NIWO_Litter_massdata_2018-08_raw.csv`). Name these datasets “Neonics” and “Litter”, respectively. Be sure to include the subcommand to read strings in as factors.

```
library(tidyverse);library(lubridate);library(here)
getwd()
```

```
## [1] "/home/guest/R/EDE_Fall2025"
```

```
Neonics = read.csv(
  file = here('Data', 'Raw', 'ECOTOX_Neonicotinoids_Insects_raw.csv'),
  stringsAsFactors = TRUE
)
```

```
Litter = read.csv(
  file = here('Data', 'Raw', 'NEON_NIWO_Litter_massdata_2018-08_raw.csv'),
  stringsAsFactors = TRUE
)
```

Learn about your system

2. The neonicotinoid dataset was collected from the Environmental Protection Agency’s ECOTOX Knowledgebase, a database for ecotoxicology research. Neonicotinoids are a class of insecticides used widely in agriculture. The dataset that has been pulled includes all studies published on insects. Why might we be interested in the ecotoxicology of neonicotinoids on insects? Feel free to do a brief internet search if you feel you need more background information.

Answer: Because while these insecticides are designed to kill pest insects, their widespread use and systemic nature pose significant threats to beneficial insects that are vital for healthy ecosystems and agriculture.

3. The Niwot Ridge litter and woody debris dataset was collected from the National Ecological Observatory Network, which collectively includes 81 aquatic and terrestrial sites across 20 ecoclimatic domains. 32 of these sites sample forest litter and woody debris, and we will focus on the Niwot Ridge long-term ecological research (LTER) station in Colorado. Why might we be interested in studying litter and woody debris that falls to the ground in forests? Feel free to do a brief internet search if you feel you need more background information.

Answer: Because they are crucial components of a healthy and functioning ecosystem. They play a vital role in nutrient cycling, soil health, and providing habitat for a wide range of organisms.

4. How is litter and woody debris sampled as part of the NEON network? Read the NEON_Litterfall_UserGuide.pdf document to learn more. List three pieces of salient information about the sampling methods here:

Answer: 1. Two different trap types are used based on debris size. 2. The sampling frequency varies by the trap type and vegetation type. 3. All samples are sorted and sub-sampled for chemical analysis.

Obtain basic summaries of your data (Neonics)

5. What are the dimensions of the dataset?

```
dim(Neonics)
```

```
## [1] 4623 30
```

6. Using the `summary` function on the “Effect” column, determine the most common effects that are studied. Why might these effects specifically be of interest? [Tip: The `sort()` command is useful for listing the values in order of magnitude...]

```
sort(summary(Neonics$Effect), decreasing = T)
```

##	Population	Mortality	Behavior	Feeding behavior
##	1803	1493	360	255
##	Reproduction	Development	Avoidance	Genetics
##	197	136	102	82
##	Enzyme(s)	Growth	Morphology	Immunological
##	62	38	22	16
##	Accumulation	Intoxication	Biochemistry	Cell(s)
##	12	12	11	9
##	Physiology	Histology	Hormone(s)	
##	7	5	1	

Answer: The most common effects are Population(1803) and Mortality(1493), because they represent the most direct and ecologically significant ways to measure the impact of an insecticide.

7. Using the `summary` function, determine the six most commonly studied species in the dataset (common name). What do these species have in common, and why might they be of interest over other insects? Feel free to do a brief internet search for more information if needed. [TIP: Explore the help on the `summary()` function, in particular the `maxsum` argument...]

```
summary(Neonics$Species.Common.Name, maxsum = 6+1)
```

##	Honey Bee	Parasitic Wasp	Buff Tailed Bumblebee
##	667	285	183
##	Carniolan Honey Bee	Bumble Bee	Italian Honeybee
##	152	140	113
##	(Other)		
##	3083		

Answer: The six most commonly studied species in the dataset are the Honey Bee, Parasitic Wasp, Buff Tailed Bumblebee, Carniolan Honey Bee, Bumble Bee, and Italian Honeybee. These species are of high interest because they provide essential ecosystem services that are vital to both agriculture and the health of natural environments.

8. Concentrations are always a numeric value. What is the class of `Conc.1..Author.` column in the dataset, and why is it not numeric? [Tip: Viewing the dataframe may be helpful...]

```
class(Neonics$Conc.1..Author.)
```

```
## [1] "factor"
```

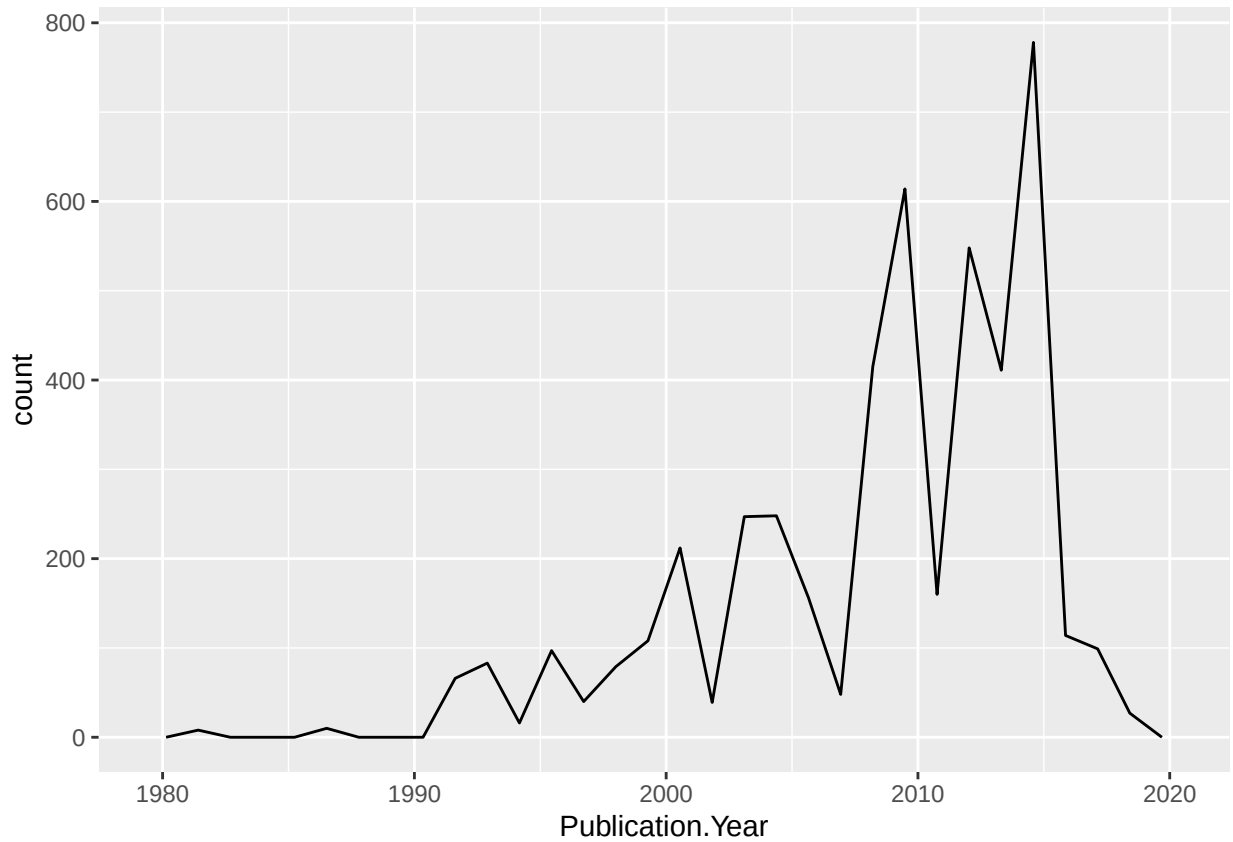
Answer: It is factor. It is not a numeric class because the column contains non-numeric values. When a data import function reads a column containing a mix of numbers and text, it defaults to classifying the entire column as character or factor to ensure no information is lost.

Explore your data graphically (Neonics)

9. Using `geom_freqpoly`, generate a plot of the number of studies conducted by publication year.

```
ggplot(Neonics) +  
  geom_freqpoly(aes(x=Publication.Year))
```

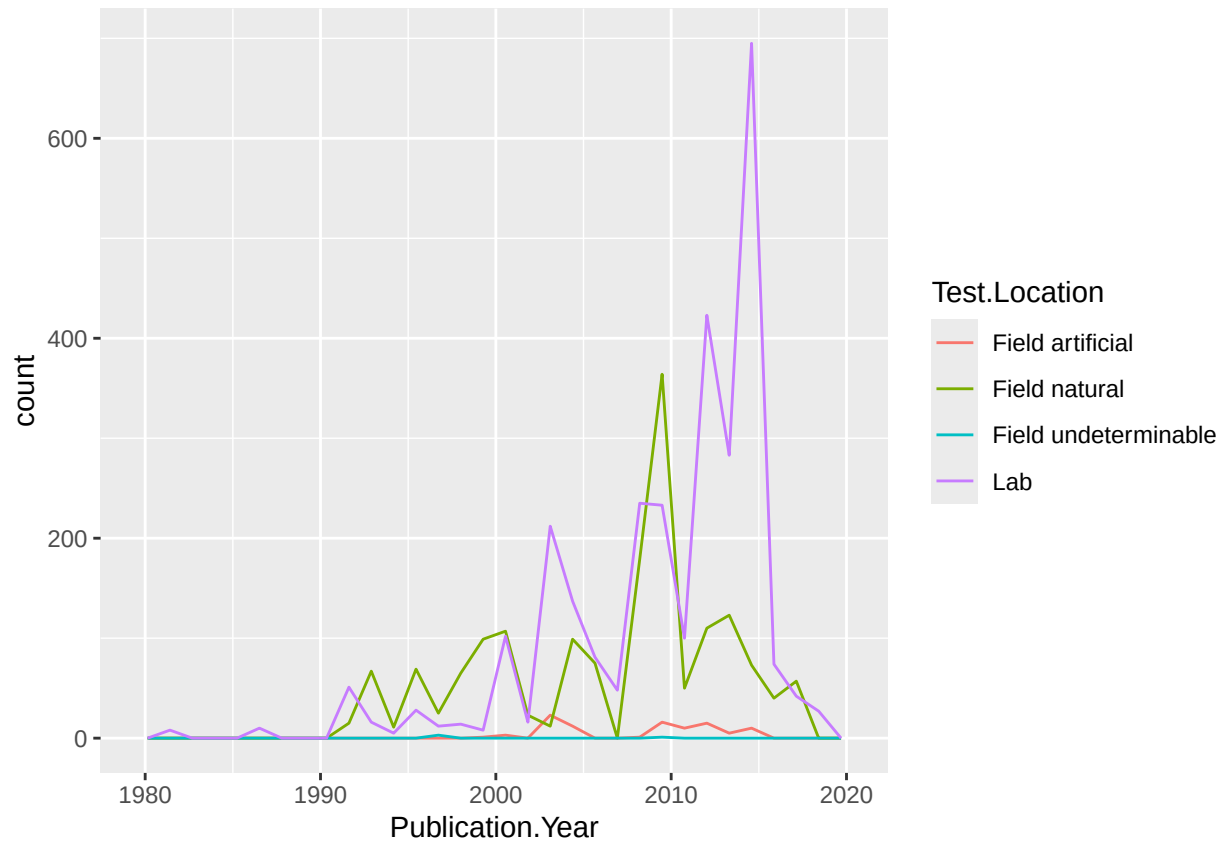
'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.



10. Reproduce the same graph but now add a color aesthetic so that different Test.Location are displayed as different colors.

```
ggplot(Neonics) +  
  geom_freqpoly(aes(x=Publication.Year, colour = Test.Location))
```

'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.



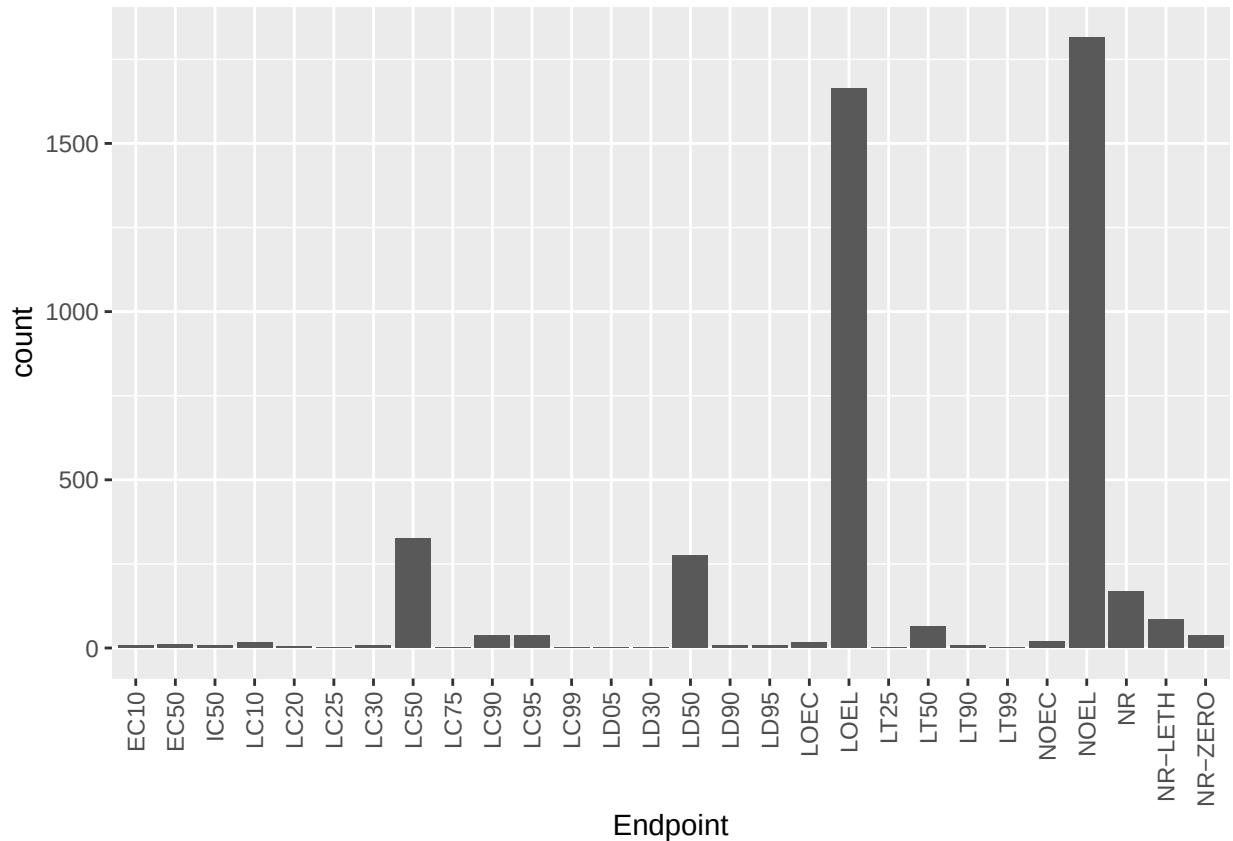
Interpret this graph. What are the most common test locations, and do they differ over time?

Answer: Lab

11. Create a bar graph of Endpoint counts. What are the two most common end points, and how are they defined? Consult the ECOTOX_CodeAppendix for more information.

[**TIP:** Add `theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))` to the end of your plot command to rotate and align the X-axis labels...]

```
ggplot(Neonics) +
  geom_bar(aes(x=Endpoint)) +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))
```



Answer: NOEL & LOEL. NOEL: No-observable-effect-level - highest dose (concentration) producing effects not significantly different from responses of controls according to author's reported statistical test (NOEL/NOEC) LOEL: Lowest-observable-effect-level - lowest dose (concentration) producing effects that were significantly different (as reported by authors) from responses of controls (LOEL/LOEC)

Explore your data (Litter)

- Determine the class of collectDate. Is it a date? If not, change to a date and confirm the new class of the variable. Using the `unique` function, determine which dates litter was sampled in August 2018.

```
class(Litter$collectDate)
```

```
## [1] "factor"
```

```
Litter$collectDate = ymd(Litter$collectDate)
class(Litter$collectDate)
```

```
## [1] "Date"
```

```
unique(Litter$collectDate)
```

```
## [1] "2018-08-02" "2018-08-30"
```

13. Using the `unique` function, list the different plotIDs sampled at Niwot Ridge. How is the information obtained from `unique` different from that obtained from `summary`?

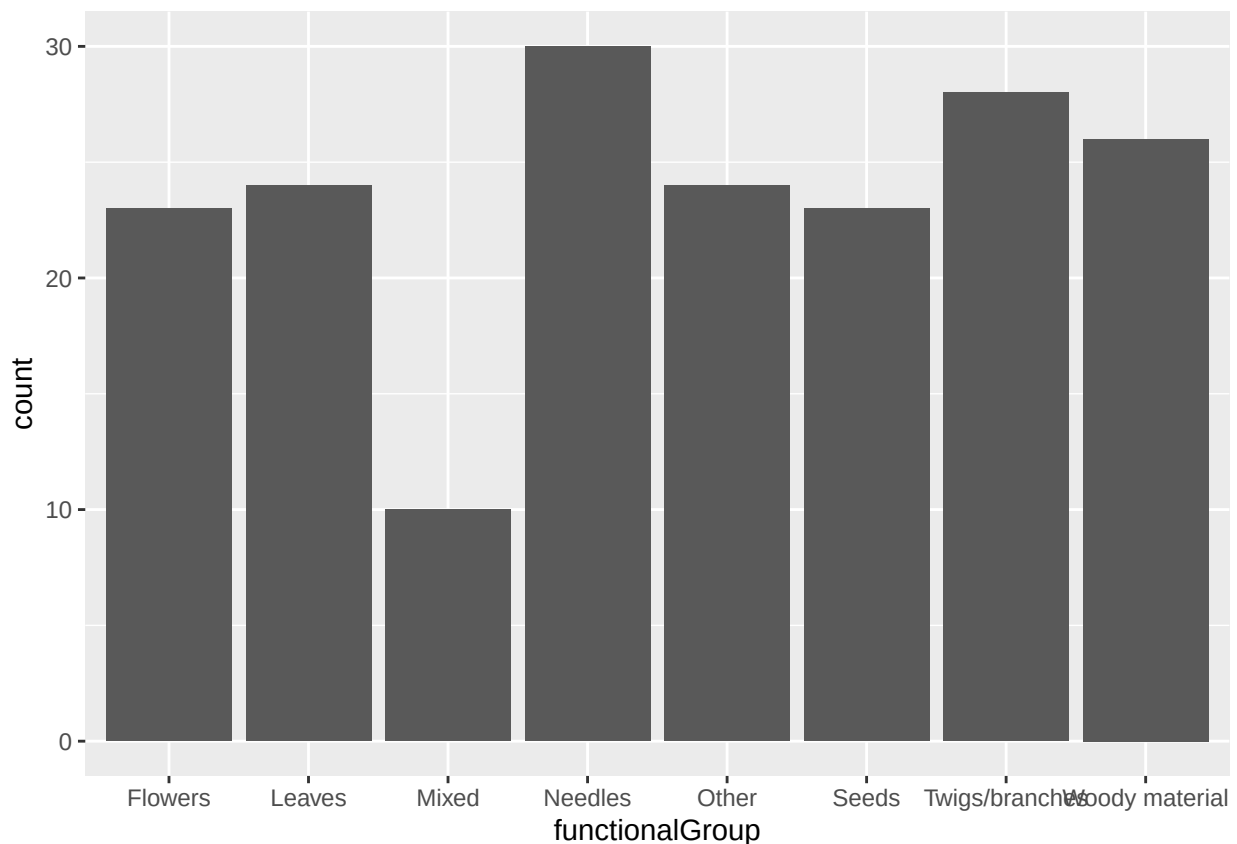
```
unique(Litter$plotID)
```

```
## [1] NIWO_061 NIWO_064 NIWO_067 NIWO_040 NIWO_041 NIWO_063 NIWO_047 NIWO_051  
## [9] NIWO_058 NIWO_046 NIWO_062 NIWO_057  
## 12 Levels: NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 ... NIWO_067
```

Answer: `unique()` tells you what values exist, while `summary()` tells you how many times each value appears.

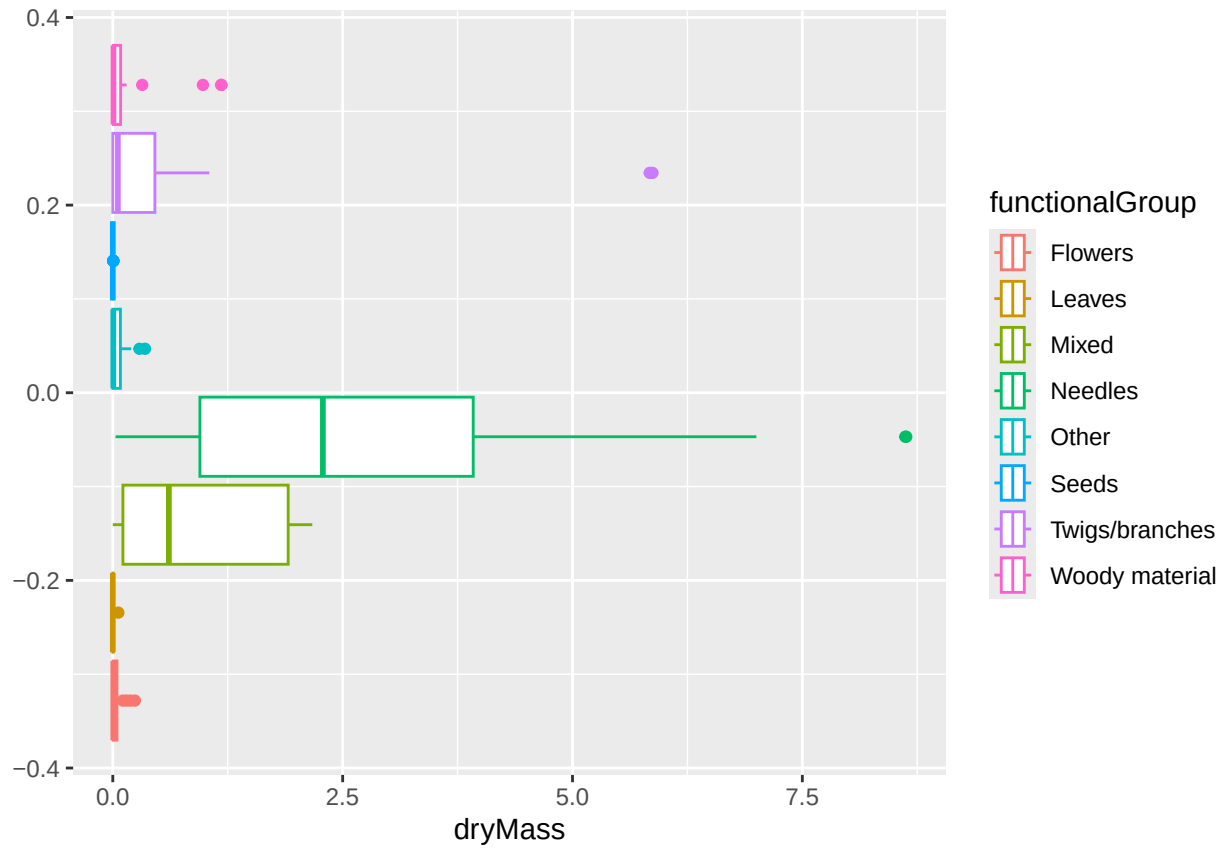
14. Create a bar graph of functionalGroup counts. This shows you what type of litter is collected at the Niwot Ridge sites. Notice that litter types are fairly equally distributed across the Niwot Ridge sites.

```
ggplot(Litter) +  
  geom_bar(aes(x=functionalGroup))
```

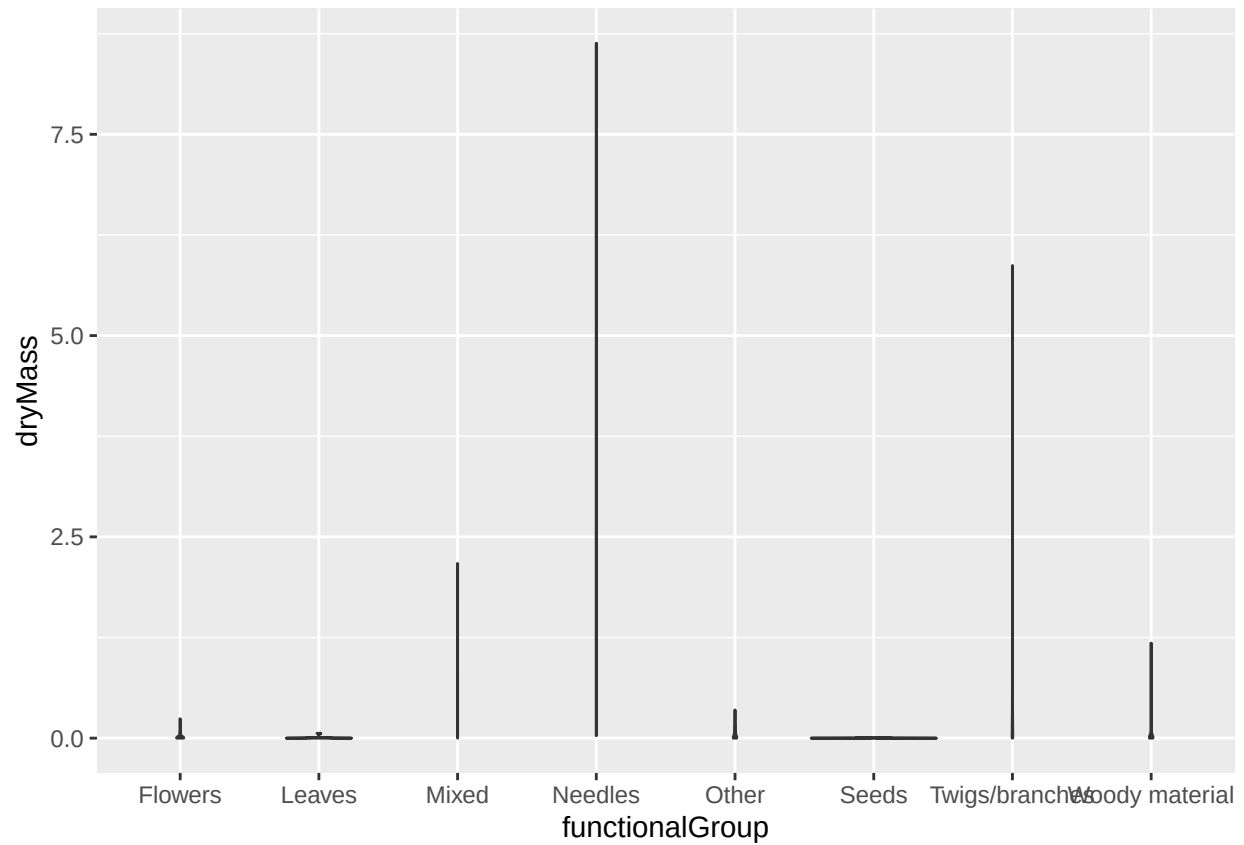


15. Using `geom_boxplot` and `geom_violin`, create a boxplot and a violin plot of `dryMass` by `functionalGroup`.

```
ggplot(Litter, aes(x = dryMass, color = functionalGroup)) +  
  geom_boxplot()
```



```
ggplot(Litter, aes(y = dryMass, x = functionalGroup)) +  
  geom_violin()
```

Why is the boxplot a more effective visualization option than the violin plot in this case?

Answer: The drymass data does not have good distribution.

What type(s) of litter tend to have the highest biomass at these sites?

Answer: Needles